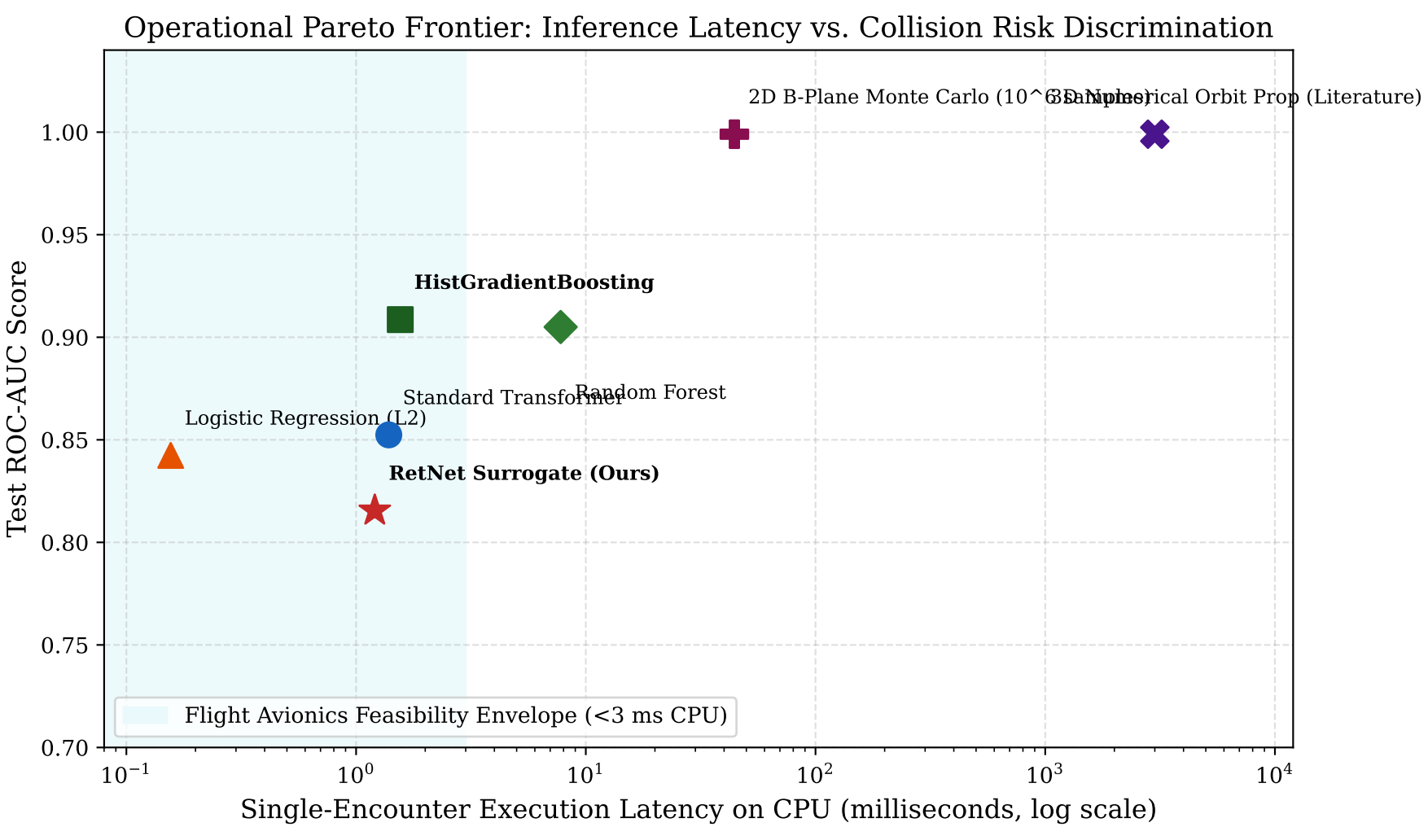
Operational Pareto Frontier: Inference Latency vs. Collision Risk Discrimination
Test ROC-AUC Score
Single-Encounter Execution Latency on CPU (milliseconds, log scale)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
2D B-Plane Monte Carlo (10^6 Samples)
3D Numerical Orbit Prop (Literature)
HistGradientBoosting
Random Forest
Standard Transformer
Logistic Regression (L2)
RetNet Surrogate (Ours)
Flight Avionics Feasibility Envelope (<3 ms CPU)